

Sourav Raxit

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Education

- 2023 – Ongoing  **Ph.D., University of New Orleans** in Computer Science.
Research Topic: *Vision Based Localization and Planning, Motion Planning in GPS Denied Environment, Robotic Information Gathering.*
- 2023 – 2025  **MS, University of New Orleans** in Computer Science.
Thesis title: *Real-time Fiducial Marker Based Localization for Autonomous Unmanned Aerial Vehicle Navigation.*
-  **Grad Certificate, University of New Orleans** in Machine Learning and AI.
- 2016 – 2021  **B.Sc. Chittagong University of Engineering and Technology** in Mechatronics and Industrial Engineering.
Thesis title: *A Comprehensive Drug Management System by Segregating Spurious and Substandard Drugs Using Blockchain Technology.*

Employment History

- Jun 2022 – Dec 2022  **Graduate Engineering Trainee**, Instrumentations Engineers Limited (Local Business Partner, Emerson)
- Aug 2023 – Ongoing  **Graduate Research Assistant**, Department of Computer Science, University of New Orleans

Research Publications

Journal Articles

- 1 **Raxit, Sourav**, A. A. R. Newaz, P. Padrao, J. Fuentes, T. Alam, and L. Bobadilla, "Dyconrig: Dynamic consensus for robotic information gathering," *Robotics and Autonomous Systems*, (Submitted).
- 2 J. Wagner, **Raxit, Sourav**, and A. A. R. Newaz, "Db-slam: Relational database-based graph simultaneous localization and mapping for resource-efficient autonomous navigation," *Intelligent Service Robotics*, (Submitted).
- 3 **Raxit, Sourav**, J. Fuentes, P. Padrao, *et al.*, "Energy-efficient multi-robot coverage path planning of non-convex regions of interests," *IEEE Robotics and Automation Letters*, pp. 1–8, 2026.  DOI: 10.1109/LRA.2026.3692093.
- 4 **Raxit, Sourav**, A. A. R. Newaz, P. Padrao, J. Fuentes, and L. Bobadilla, "Bow: Bayesian optimization over windows for motion planning in complex environments," *IEEE Robotics and Automation Letters*, vol. 10, no. 10, pp. 10 714–10 721, 2025.  DOI: 10.1109/LRA.2025.3604738.

Conference Proceedings

- 1 **Raxit, Sourav**, A. A. R. Newaz, J. Fuentes, P. Padrao, and L. Bobadilla, "Bowconnect: Parallel bayesian optimization over windows with learned local cost maps for sample-efficient kinodynamic motion planning," in *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, (Submitted), IEEE, 2026.
- 2 **Raxit, Sourav**, A. A. R. Newaz, J. Fuentes, P. Padrao, A. Cavalcanti, and L. Bobadilla, "Multi-robot trajectory planning via constrained bayesian optimization and local cost map learning with stl-based conflict resolution," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, Accepted for publication, 2026, pp. 1–8.

- 3 R. Ibn Momin, R. K. Das, **Raxit,Sourav**, U. S. Muna, and A. A. Miraj, "Design and fabrication of bengali braille embosser using solenoid," in *2024 IEEE 9th International Conference for Convergence in Technology*, Apr. 2024.
- 4 **Raxit,Sourav**, S. B. Singh, and A. A. R. Newaz, "Yolotag: Vision-based robust uav navigation with fiducial markers," in *2024 33rd IEEE International Conference on Robot and Human Interactive Communication (ROMAN)*, IEEE, Aug. 2024, pp. 311–316.
- 5 J. Hossain Gourob, **Raxit,Sourav**, and A. Hasan, "A robotic hand: Controlled with vision based hand gesture recognition system," in *2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI)*, 2021, pp. 1–4.  DOI: 10.1109/ACMI53878.2021.9528192.
- 6 **Raxit,Sourav**, S. M. Arman, and S. Banik, "Automated car parking system," in *International Conference on Mechanical Engineering and Renewable Energy 2021 (ICMERE 2021)*, Dec. 2021.
- 7 **Raxit,Sourav**, J. Hossain Gourob, and H. Kabir, "A comprehensive drug management system by segregating spurious and substandard drugs using blockchain technology," in *2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI)*, 2021, pp. 1–5.  DOI: 10.1109/ACMI53878.2021.9528238.

Skills

Languages	Strong reading, writing and speaking competencies for English, Bangla
Coding	Python, R, $\LaTeX$
Operating System	ROS1, ROS2, Linux, Windows
Simulation Software	CoppeliaSim, Gazebo, Ansys, Preoteus, Opensim, CARLA
Machine Learning Platform	Tensorflow, PyTorch
Development Board	Arduino, Raspberry Pie, STM32, Jetson.
Blockchain Technology	Hyperledger Fabric, Ethereum.
Misc.	Academic research, Teaching, Training, Consultation

Miscellaneous Experience

Awards and Achievements

- 2026  **3rd Place**, Poster Presentation, Louisiana Biomedical Research Network.
- 2023  **Early Bird Graduate Scholarship**, A scholarship Which supports graduate students.
- 2018  **Runner's Up**, BIT Arena, A robotic footbot competition in North South University.
- 2017  **Runner's Up**, A Robo car race competition arranged by Student's Union in CUET.
- 2016-2021  **Compensatory Scholarship**, Chittagong University of Engineering and Technology (CUET), Chittagong
- 2013,2015  **The Daily Prothom Alo Award**, (GPA-5 Holder in both SSC 2013 and HSC 2015

Certification

- 2022  **Training on Control Valve**. Awarded by Emerson, Singapore
- 2019  **Certified Cyber Security Specialist**. Awarded by Information and Communication Technology Division,Bangladesh
- 2018  **PLC Training for Industrial Automation**. Awarded by Institute of Energy Technology, CUET

Extra-Curricular Activities

- Aug 2019 - Sep 2020  **President**, Robo Mechatronics Association, Bangladesh (RMA), CUET
Oct 2019-Oct 2020  **Technical Head**, Radio CUET , CUET
- Sep 2019- Nov 2020  **President**, Gazipur District Student's Forum, CUET

Industrial Training

- 2018  (30 Days) S.Alam Cold Rolled Steel LTD
  (02 Days) Summit Power Ltd
- 2017  (07 Days) Bangladesh Steel Re-Rolling Mills Ltd (BSRM)
- 2016  (02 Days) Nur Mohammad Plastics

References

Prof. Abdullah Al Redwan Newaz

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Prof. Tamjidul Hoque

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